1) a)

Three real roots

One real root and two complex roots

1) b)

Four real roots

Two real roots and two complex roots

Four complex roots

1) c)

Five real roots

Three real roots and two complex roots

One real root and four complex roots

2)

Two roots are $1 + 5\text{i}$ and $1 - 5\text{i}$ which gives the quadratic: $z^2 - 2z + 26$

$z^3 - 4z^2 + 30z - 52 = (z^2 - 2z + 26)(z - 2)$

So the three roots are $1 + 5\text{i}$, $1 - 5\text{i}$ and 2

3)

Two roots are $3 - 5\text{i}$ and $3 + 5\text{i}$ which gives the quadratic: $z^2 - 6z + 34$

$-z^3 + 11z^2 - 64z + 170 = (z^2 - 6z + 34)(-z + 5)$

So the three roots are $3 - 5\text{i}, 3 + 5\text{i}$ and 5

4)

The roots are 4, $2 - 8\text{i}$ and $2 + 8\text{i}$

$\big(z - (2 - 8\text{i})\big)\big(z - (2 + 8\text{i})\big) = z^2 - 4z + 68$

$(z - 4)(z^2 - 4z + 68) = z^3 - 4z^2 + 68z - 4z^2 + 16z - 272$

So the polynomial is $z^3 - 8z^2 + 84z - 272$

$\therefore a = -8$, $b = 84$ and $c = -272$

5)

The roots are -3, $1+3\text{i}$ and $1-3\text{i}$

$\big(z-(1+3\text{i})\big)\big(z-(1-3\text{i})\big)=z^2-2z+10$

$(z+3)(z^2-2z+10)=z^3-2z^2+10z+3z^2-6z+30$

So the equation is $z^3+z^2+4z+30=0$

6) a)

$1-4\text{i}$ and 1

6) b)

$(z-1)\big(z-(1+4\text{i})\big)\big(z-(1-4\text{i})\big)$

$=(z-1)(z^2-2z+17)$

$=z^3-2z^2+17z-z^2+2z-17$

$=z^3-3z^2+19z-17$

7)

The roots are $1+\text{i}$, $1-\text{i}$, -2 and 3

$\big(z-(1+\text{i})\big)\big(z-(1-\text{i})\big)=z^2-2z+2$

$(z+2)(z-3)=z^2-z-6$

$(z^2-2z+2\,)(z^2-z-6)$

$=z^4-2z^3+2z^2-z^3+2z^2-2z-6z^2+12z-12$

$=z^4-3z^3-2z^2+10z-12$

$\therefore a=-3,\ b=-2,\ c=10,\ d=-12$

8)

The roots are $1-5\text{i}$, $1+5\text{i}$, $2+\text{i}$ and $2-\text{i}$

$\big(z-(1+5\text{i}\,)\big)\big(z-(1-5\text{i})\big)=z^2-2z+26$

$\big(z-(2+\text{i})\big)\big(z-(2-\text{i})\big)=z^2-4z+5$

$(z^2-2z+26)(z^2-4z+5)=z^4-6z^3+39z^2-114z+130$

So the equation is $\text{f}(z)=z^4-6z^3+39z^2-114z+130$

$a=1,\ b=-6,\ c=39,\ d=130$

9)

The roots are $2+\mathrm{i}$, $2-\mathrm{i}$, $4-3\mathrm{i}$ and $4+3\mathrm{i}$

$\big(z-(2+\mathrm{i})\big)\big(z-(2-\mathrm{i})\big)=z^2-4z+5$

$\big(z-(4-3i)\big)\big(z-(4+3i)\big)=z^2-8z+25$

$(z^2-4z+5\,)(z^2-8z+25)$

$=z^4-8z^3+25z^2-4z^3+32z^2-100z+5z^2-40z+125$

$=z^4-12z^3+62z^2-140z+125$

$\therefore a=-12,\ b=62,\ c=-140,\ d=125$

10)

One root is 9 so that $(z-9)$ is a factor

Therefore by inspection:

$z^3-13z^2+56z-180=(z-9)(z^2-4z+20)$

$z^2-4z+20=0$ has roots $z=\frac{4\pm\sqrt{16-80}}{2}=2\pm 4\mathrm{i}$

So the three roots are $9, 2+4\mathrm{i}$ and $2-4\mathrm{i}$

11)

$(z-4)$ is a factor

Therefore by inspection:

$z^3-10z^2+34z-40=(z-4)(z^2-6z+10)$

$z^2-6z+10$ has roots $z=\frac{6\pm\sqrt{36-40}}{2}=3\pm\mathrm{i}$

So the three roots are $4,\ 3+\mathrm{i}$ and $3-\mathrm{i}$

12)

$(z+3)$ is a factor so -3 is a solution

Therefore by inspection:

$z^3+z^2+11z+51=(z+3)(z^2-2z+17)$

$z^2-2z+17=0$ has solutions $z=\frac{2\pm\sqrt{4-68}}{2}=1\pm 4\mathrm{i}$

So the three solutions are $-3,\ 1+4\mathrm{i}$ and $1-4\mathrm{i}$

13)

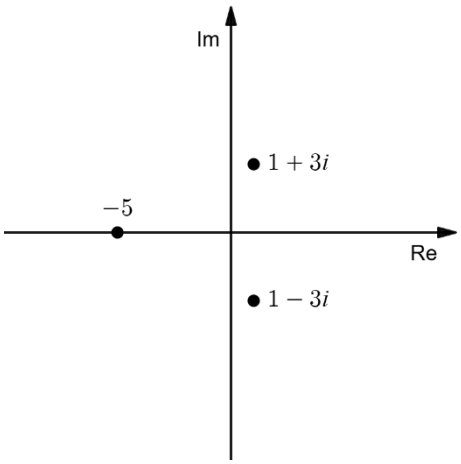
Two of the roots are $1-3\mathrm{i}$ and $1+3\mathrm{i}$

$\big(z-(1-3\mathrm{i})\big)\big(z-(1+3\mathrm{i})\big)=z^2-2z+10$

By inspection:

$(z^2-2z+10\,)(z+5)=z^3+3z^2+50$

So the three roots are -5, $1-3\mathrm{i}$ and $1+3\mathrm{i}$



Area $=\frac{1}{2}\times 6\times 6=18$ units^2

14)

One of the roots is 3

By inspection:

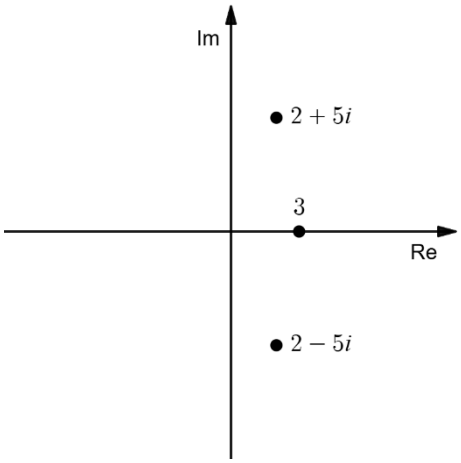
$(z-3)(z^2-4z+29)=z^3-7z^2+41z-87$

And solving $z^2-4z+29=0$ gives

$z^2-4z+29=0$

$(z-2)^2-4=-29\Rightarrow z-2=\pm\sqrt{-25}\Rightarrow z=2\pm 5\mathrm{i}$

So the three roots are 3, $2+5\mathrm{i}$ and $2-5\mathrm{i}$



Area $=\frac{1}{2}\times 1\times 10=5$ units^2

15) a)

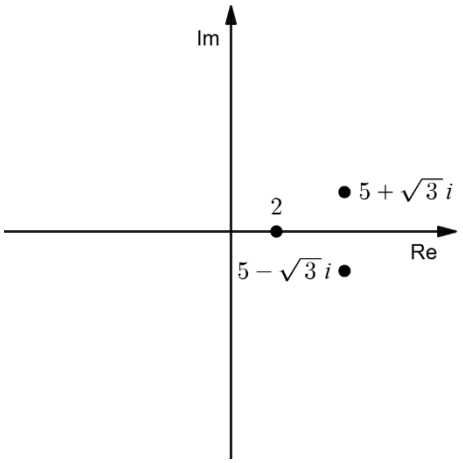
Two of the roots are $5-\sqrt{3}\mathrm{i}$ and $5+\sqrt{3}\mathrm{i}$

$\left(z-\left(5-\sqrt{3}\mathrm{i}\right)\right)\left(z-\left(5+\sqrt{3}\mathrm{i}\right)\right)=z^2-10z+28$

By inspection:

$(z^2-10z+28\,)(z-2)=z^3-12z^2+48z-56$

So the three roots are 2, $5-\sqrt{3}\mathrm{i}$ and $5+\sqrt{3}\mathrm{i}$



Area $=\frac{1}{2}\times 2\sqrt{3}\times 3=3\sqrt{3}$ units^2

15) b)

Calculate lengths of sides:

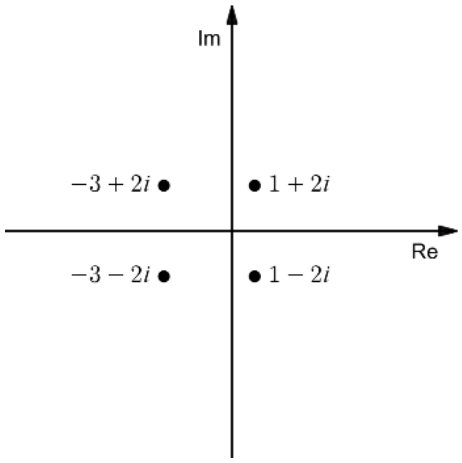
Between $5+\sqrt{3}\mathrm{i}$ and $5-\sqrt{3}\mathrm{i}$ it is $2\sqrt{3}$

Between 2 and $5+\sqrt{3}\mathrm{i}$ it is $\sqrt{3^2+\left(\sqrt{3}\right)^2}=\sqrt{12}=2\sqrt{3}$

Between 2 and $5-\sqrt{3}\mathrm{i}$ it is $\sqrt{3^2+\left(-\sqrt{3}\right)^2}=\sqrt{12}=2\sqrt{3}$

Since all the lengths are the same it must be an equilateral triangle.

16) a)



16) b)

$(z-(1+2\mathrm{i}))(z-(1-2\mathrm{i}))(z-(-3+2\mathrm{i}))(z-(-3-2\mathrm{i}))$
$=(z^2-2z+5)(z^2+6z+13)$
$=z^4+6z^3+13z^2-2z^3-12z^2-26z+5z^2+30z+65$
$=z^4+4z^3+26z^2+4z+65$
$\therefore \mathrm{f}(z)=z^4+4z^3+26z^2+4z+65$

17) a)

$3+5\mathrm{i}$, $3-5\mathrm{i}$ and 3

17) b)

$(z-(3+5\mathrm{i}))(z-(3-5\mathrm{i}))(z-3)(z-3)$
$=(z^2-6z+34)(z^2-6z+9)$
$=z^4-6z^3+9z^2-6z^3+36z^2-54z+34z^2-204z+306$
$=z^4-12z^3+79z^2-258z+306$
$\therefore \mathrm{f}(z)=z^4-12z^3+79z^2-258z+306$

18) a)

$(2+4\mathrm{i})^3=2^3+3\times2^2\times4\mathrm{i}+3\times2\times(4\mathrm{i})^2+\times(4\mathrm{i})^3$
$=8+48\mathrm{i}-96-64\mathrm{i}$
$=-88-16\mathrm{i}$

18) b)

$(2+4\mathrm{i})^3 - 7(2+4\mathrm{i})^2 + 32(2+4\mathrm{i}) - 60$

$= -88 - 16\mathrm{i} - 7(-12+16\mathrm{i}) + 64 + 128\mathrm{i} - 60$

$= 0$ ∴ by the factor theorem, $2+4\mathrm{i}$ is a root of $\mathrm{f}(z) = 0$

18) c)

Two of the roots are $2+4\mathrm{i}$ and $2-4\mathrm{i}$

$\big(z-(2+4\mathrm{i})\big)\big(z-(2-4\mathrm{i})\big) = z^2 - 4z + 20$

By inspection:

$z^3 - 7z^2 + 32z - 60 = (z^2 - 4z + 20)(z-3)$

19) a)

$(1+\mathrm{i})^4 - (1+\mathrm{i})^3 - 6(1+\mathrm{i})^2 + 14(1+\mathrm{i}) - 12$

$= 2\mathrm{i} \times 2\mathrm{i} - 2\mathrm{i}(1+\mathrm{i}) - 6 \times 2\mathrm{i} + 14 + 14\mathrm{i} - 12$

$= -4 - 2\mathrm{i} + 2 - 12\mathrm{i} + 14 + 14\mathrm{i} - 12$

$= 0$

∴ $1+\mathrm{i}$ is a root.

19) b)

Two of the roots are $1+\mathrm{i}$ and $1-\mathrm{i}$

$\big(z-(1+\mathrm{i})\big)\big(z-(1-\mathrm{i})\big) = z^2 - 2z + 2$

	z^2	$+z$	-6
z^2	z^4	$+z^3$	$-6z^2$
$-2z$	$-2z^3$	$-2z^2$	$+12z$
$+2$	$+2z^2$	$+2z$	-12

And

$z^2 + z - 6 = (z-2)(z+3) = 0$ has solutions $z = 2$ and $z = -3$

∴ the roots of $z^4 - z^3 - 6z^2 + 14z - 12 = 0$ are $2, -3, 1+\mathrm{i}$ and $1-\mathrm{i}$

20)

Two of the roots are $2-7\mathrm{i}$ and $2+7\mathrm{i}$

$\big(z-(2-7\mathrm{i})\big)\big(z-(2+7\mathrm{i})\big)=z^2-4z+53$

	z^2	$-2z$	$+10$
z^2	z^4	$-2z^3$	$+10z^2$
$-4z$	$-4z^3$	$+8z^2$	$-40z$
$+53$	$+53z^2$	$-106z$	$+530$

And

$z^2-2z+10=0$ has solutions $z=1-3\mathrm{i}$ and $z=1+3\mathrm{i}$

$\therefore$ the roots of $z^4-6z^3+71z^2-146z+530=0$ are $2-7\mathrm{i}, 2+7\mathrm{i}, 1-3\mathrm{i}$ and $1+3\mathrm{i}$

21) a)

Two of the solutions are $1+3\mathrm{i}$ and $1-3\mathrm{i}$

$\big(z-(1+3\mathrm{i})\big)\big(z-(1-3\mathrm{i})\big)=z^2-2z+10$

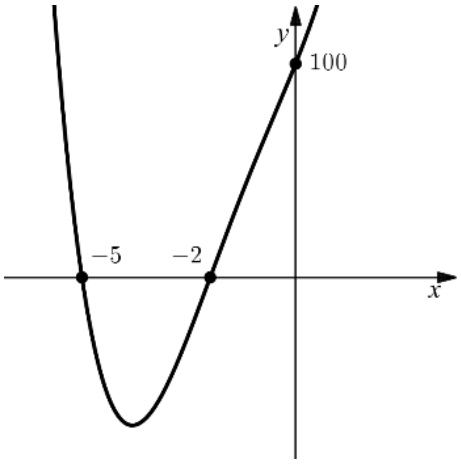
	z^2	$+7z$	$+10$
z^2	z^4	$+7z^3$	$+10z^2$
$-2z$	$-2z^3$	$-14z^2$	$-20x$
$+10$	$+10z^2$	$+70z$	$+100$

And

$z^2+7z+10=(z+5)(z+2)=0$ has solutions $z=-5$ and $z=-2$

$\therefore$ the solutions of $z^4+5z^3+6z^2+50z+100=0$ are $-5, -2, 1+3\mathrm{i}$ and $1-3\mathrm{i}$

21) b)



22) a)

$(1+2\mathrm{i})^4+4(1+2\mathrm{i})^3+3(1+2\mathrm{i})^2+10(1+2\mathrm{i})+50$

$=(-3+4\mathrm{i})^2+4(-3+4\mathrm{i})(1+2\mathrm{i})+3(-3+4\mathrm{i})+10(1+2\mathrm{i})+50$

$=-7-24\mathrm{i}-44-8\mathrm{i}-9+12\mathrm{i}+10+20\mathrm{i}+50$

$=0$

$\therefore 1+2\mathrm{i}$ is a root

22) b)

Two of the solutions are $1+2\mathrm{i}$ and $1-2\mathrm{i}$

$\big(z-(1+2\mathrm{i})\big)\big(z-(1-2\mathrm{i})\big)=z^2-2z+5$

	z^2	$+6z$	$+10$
z^2	z^4	$+6z^3$	$+10z^2$
$-2z$	$-2z^3$	$-12z^2$	$-20z$
$+5$	$+5z^2$	$+30z$	$+50z$

And

$z^2+6z+10=0$ has solutions $z=-3+\mathrm{i}$ and $z=-3-\mathrm{i}$

$\therefore$ the roots of $z^4+4z^3+3z^2+10z+50=0$ are $1+2\mathrm{i},\ 1-2\mathrm{i}, -3+\mathrm{i}$ and $-3-\mathrm{i}$

22) c)

All of the solutions of $\mathrm{f}(x)=0$ are complex and so the graph of $y=\mathrm{f}(x)$ will not intersect the x-axis.

23) a)

Using factor theorem:

$\mathrm{f}(z)=z^3-13z^2+56z-78$

$\mathrm{f}(1)=1-13+56-78\neq 0$

$\mathrm{f}(2)=8-52+112-78\neq 0$

$\mathrm{f}(3)=27-117+168-78=0\ \therefore (z-3)$ is a factor

$z^3-13z^2+56z-78=(z-3)(z^2-10z+26)$

$z=\frac{10\pm\sqrt{100-104}}{2}$

So the solutions are $z=3$, $z=5-\mathrm{i}$ and $z=5+\mathrm{i}$

23) b)

Using factor theorem:

$f(z) = z^3 - 6z^2 + 15z - 14$

$f(1) = 1 - 6 + 15 - 14 \neq 0$

$f(2) = 8 - 24 + 30 - 14 = 0 \therefore (z - 2)$ is a factor

$z^3 - 6z^2 + 15z - 14 = (z - 2)(z^2 - 4z + 7)$

$z = \frac{4 \pm \sqrt{16-28}}{2}$

So the solutions are $z = 2$, $z = 2 - \sqrt{3}\mathrm{i}$ and $z = 2 + \sqrt{3}\mathrm{i}$

23) c)

Using factor theorem:

$f(z) = z^3 - 2z^2 - 2z + 12$

$f(1) = 1 - 2 - 2 + 12 \neq 0$

$f(2) = 8 - 8 - 4 + 12 \neq 0$

$f(-2) = -8 - 8 + 4 + 12 \therefore (z + 2)$ is a factor

$z^3 - 2z^2 - 2z + 12 = (z + 2)(z^2 - 4z + 6)$

$z = \frac{4 \pm \sqrt{16-24}}{2}$

So the solutions are $z = -2$, $z = 2 - \sqrt{2}\mathrm{i}$ and $z = 2 + \sqrt{2}\mathrm{i}$

23) d)

Using factor theorem:

$f(z) = 3z^3 - 20z^2 + 54z - 28$

$f(1) = 3 - 20 + 54 - 28 \neq 0$

$f\left(\frac{1}{3}\right) = \frac{1}{9} - \frac{20}{9} + 18 - 28 \neq 0$

$f(2) = 24 - 80 + 108 - 28 \neq 0$

$f\left(\frac{2}{3}\right) = \frac{8}{9} - \frac{80}{9} + 36 - 28 = 0 \ \therefore (3z - 2)$ is a factor

$3z^3 - 20z^2 + 54z - 28 = (3z - 2)(z^2 - 6z + 14)$

$z = \frac{6 \pm \sqrt{36-56}}{2}$

So the solutions are $z = \frac{2}{3}$, $z = 3 - \sqrt{5}\mathrm{i}$ and $z = 3 + \sqrt{5}\mathrm{i}$

24) a)

Since complex roots for polynomials with real coefficients are conjugate pairs, the other roots must be $1-4\mathrm{i}$ and 1 (since they have to be in a straight line)

$\therefore 1+4\mathrm{i},\ 1-4\mathrm{i}$ and 1

24) b)

$\big(z-(1-4\mathrm{i})\big)\big(z-(1+4\mathrm{i})\big)(z-1)$

$=(z^2-2z+17)(z-1)$

$=z^3-2z^2+17z-z^2+2z-17$

$=z^3-3z^2+19z-17$

$\therefore \mathrm{f}(z)=z^3-3z^2+19z-17$

25) a)

Since complex roots for polynomials with real coefficients are conjugate pairs, the roots must be $2-3\mathrm{i}, 2+2\mathrm{i}$ and 2 (since they have to be in a straight line)

$\therefore 2-3\mathrm{i},\ 2+3\mathrm{i},\ 2+2\mathrm{i},\ 2-2\mathrm{i}$ and 2

25) b)

$\big(z-(2-3\mathrm{i})\big)\big(z-(2+3\mathrm{i})\big)\big(z-(2+2\mathrm{i})\big)\big(z-(2-2\mathrm{i})\big)(z-2)$

$=(z^2-4z+13)(z^2-4z+8)(z-2)$

$=(z^4-4z^3+8z^2-4z^3+16z^2-32z+13z^2-52z+104)(z-2)$

$=(z^4-8z^3+37z^2-84z+104)(z-2)$

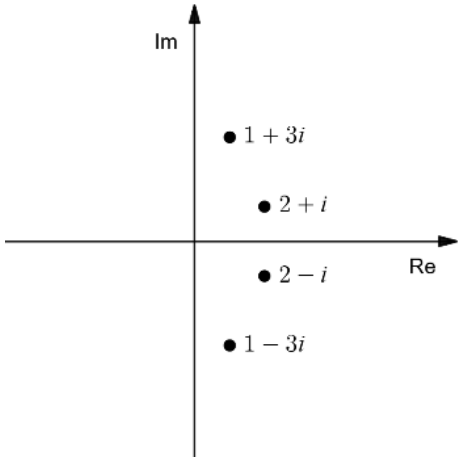
$=z^5-8z^4+37z^3-84z^2+104z-2z^4+16z^3-74z^2+168z-208$

$=z^5-10z^4+53z^3-158z^2+272z-208$

$\therefore \mathrm{f}(z)=z^5-10z^4+53z^3-158z^2+272z-208$

26) a)

The roots of the quartic are $2 + \mathrm{i}$, $2 - \mathrm{i}$, $1 - 3\mathrm{i}$ and $1 + 3\mathrm{i}$



26) b)

Area is $\frac{1}{2} \times 1 \times (2 + 6) = 4$ units^2

26) c)

$\big(z - (2 + \mathrm{i})\big)\big(z - (2 - \mathrm{i})\big) = z^2 - 4z + 5$

$\big(z - (1 - 3\mathrm{i})\big)\big(z - (1 + 3\mathrm{i})\big) = z^2 - 2z + 10$

$(z^2 - 4z + 5)(z^2 - 2z + 10) = z^4 - 2z^3 + 10z^2 - 4z^3 + 8z^2 - 40z + 5z^2 - 10z + 50$

$= z^4 - 6z^3 + 23z^2 - 50z + 50$

$\therefore a = -6$, $b = 23$, $c = -50$ and $d = 50$

27) a)

$(1 + 2\mathrm{i})(1 + 2\mathrm{i}) = 1 + 4\mathrm{i} - 4 = -3 + 4\mathrm{i}$

27) b)

$\big(z - (1 + 2\mathrm{i})\big)\big(z - (1 - 2\mathrm{i})\big)\big(z - (-3 + 4\mathrm{i})\big)\big(z - (-3 - 4\mathrm{i})\big)$

$= (z^2 - 2z + 5)(z^2 + 6z + 25)$

$= z^4 + 6z^3 + 25z^2 - 2z^3 - 12z^2 - 50z + 5z^2 + 30z + 125$

$= z^4 + 4z^3 + 18z^2 - 20z + 125$

28) a)

Since $1-2\text{i}$ is a root $1+2\text{i}$ is also a root

$\big(z-(1-2\text{i})\big)\big(z-(1+2\text{i})\big)=z^2-2z+5$

$\therefore$ by inspection

$z^3-4z^2+kz-10=(z-2)(z^2-2z+5)$

So that the roots are $1-2\text{i}$, $1+2\text{i}$ and 2

28) b)

$(z-2)(z^2-2z+5)=z^3-4z^2+9z-10$

$\therefore k=9$

29) a)

Let $\text{f}(z)=z^3+kz^2-5z-12$

Since $z-4$ is a factor:

$\text{f}(4)=4^3+k(4)^2-5(4)-12=0$

$64+16k-20-12=0$

$\therefore k=-2$

29) b)

$z^3-2z^2-5z-12=(z-4)(z^2+2z+3)$

So that the roots are 4, $-1-\sqrt{2}\text{i}$ and $-1+\sqrt{2}\text{i}$

30) a)

Since $3-\text{i}$ is a root, then $3+\text{i}$ is also a root:

$\big(z-(3-\text{i})\big)\big(z-(3+\text{i})\big)=z^2-6z+10$

	$2z$	-7
z^2	$2z^3$	$-7z^2$
$-6z$	$-12z^2$	$+42z$
$+10$	$+20z$	-70

So that $20+42=p$ and $q=-70$

$\therefore p=62$ and $q=-70$

30) b)

By inspection:

$z^3 - 19z^2 + 62z - 70 = (z^2 - 6z + 10)(2z - 7)$

So that the roots are $\frac{7}{2}$, $3 - \mathrm{i}$ and $3 + \mathrm{i}$

31) a)

Two of the solutions are $2 - \mathrm{i}$ and $2 + \mathrm{i}$

$\big(z - (2 - \mathrm{i})\big)\big(z - (2 + \mathrm{i})\big) = z^2 - 4z + 5$

$z^4 - 6z^3 + pz^2 + qz + 20 = (z^2 - 4z + 5)(z^2 - 2z + 4)$

31) b)

$p = 4 + 8 + 5 = 17$

$q = -16 - 10 = -26$

32) a)

Since $2\mathrm{i}$ is a root, then $-2\mathrm{i}$ is also a root:

$(z - 2\mathrm{i})(z - -2\mathrm{i}) = z^2 + 4$

	z^2	$-3z$	-10
z^2	z^4	$-3z^3$	$-10z^2$
$+0z$	$+0z^3$	$-0z^2$	$-0z$
$+4$	$+4z^2$	$-12z$	-40

$\therefore p = -12$ and $q = -40$

32) b)

$z^2 - 3z - 10 = 0 = (z + 2)(z - 5)$ has roots $z = -2$ and $z = 5$

The roots are $2\mathrm{i}$, $-2\mathrm{i}$, -2 and 5

33) a)

Let the root be $a\mathrm{i}$ for some real value a

Then

$(a\mathrm{i})^4 - 6(a\mathrm{i})^3 + 2(a\mathrm{i})^2 - 54(a\mathrm{i}) - 63 = 0$

$a^4 + 6a^3\mathrm{i} - 2a^2 - 54a\mathrm{i} - 63 = 0$

$(a^4 - 2a^2 - 63) + (6a^3 - 54a)\mathrm{i} = 0$

Real part: $a^4 - 2a^2 - 63 = 0 \Rightarrow (a^2 - 3)(a^2 + 7) = 0$ so $a = \pm 3$ or $a = \pm\sqrt{7}\mathrm{i}$

Since a is real $a = \pm 3$

So the purely imaginary roots are $3\mathrm{i}$ and $-3\mathrm{i}$

33) b)

$(z - 3\mathrm{i})(z + 3\mathrm{i}) = z^2 + 9$

$z^4 - 6z^3 + 2z^2 - 54z - 63 = 0$

$(z^2 + 9)(z^2 - 6z - 7) = 0$

$(z^2 + 9)(z - 7)(z + 1) = 0$

So the roots are

$z = 7$, $z = -1$, $z = 3\mathrm{i}$ and $z = -3\mathrm{i}$

34)

Since $3 - 2\mathrm{i}$ is a root, then $3 + 2\mathrm{i}$ is also a root:

$\big(z - (3 - 2\mathrm{i})\big)\big(z - (3 + 2\mathrm{i})\big) = z^2 - 6z + 13$

By inspection:

$z^4 + az^3 + bz^2 + az + 13 = (z^2 - 6z + 13)(z^2 + Pz + 1)$

$= z^4 + Pz^3 + z^2 - 6z^3 - 6Pz^2 - 6z + z^2 - 6z + 13$

$= z^4 + (P - 6)z^3 + (2 - 6P)z^2 - 6z + 13$

Comparing linear terms: $a = -6$

Comparing cubic terms: $P - 6 = -6$ so $P = 0$

So

$z^4 + az^3 + bz^2 + az + 13 = (z^2 - 6z + 13)(z^2 + 1)$

35) a)

Since $-1+\frac{\sqrt{3}}{2}\mathrm{i}$ is a root, $-1-\frac{\sqrt{3}}{2}\mathrm{i}$ is also a root:

$4\left(z-\left(-1+\frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z-\left(-1-\frac{\sqrt{3}}{2}\mathrm{i}\right)\right)=4\left(z^2+2z+\frac{7}{4}\right)=4z^2+8z+7$

$4z^4+16z^3+51z^2+kz+49=(4z^2+8z+7)(z^2+2z+7)$

So $-1+\sqrt{6}\mathrm{i}$ and $-1-\sqrt{6}\mathrm{i}$ are also roots.

Therefore, the roots are $-1+\frac{\sqrt{3}}{2}\mathrm{i}$, $-1-\frac{\sqrt{3}}{2}\mathrm{i}$, $-1+\sqrt{6}\mathrm{i}$ and $-1-\sqrt{6}\mathrm{i}$

35) b)

$k=56+14=70$

36) a)

$(2-\mathrm{i})(2-\mathrm{i})=4-4\mathrm{i}-1=3-4\mathrm{i}$

36) b)

$(3-4\mathrm{i})(2-\mathrm{i})=6-8\mathrm{i}-3\mathrm{i}-4=2-11\mathrm{i}$

36) c)

$p(2-11\mathrm{i})+5(3-4\mathrm{i})+12(2-\mathrm{i})+q=0$

$2p-11p\mathrm{i}+15-20\mathrm{i}+24-12\mathrm{i}+q=0$

$2p+39+q+(-23p-32)\mathrm{i}=0$

Imaginary part: $-23p-32=0$ so $p=-\frac{32}{11}$

Real part: $2\times\frac{20}{23}+39+q=0$ so $q=-\frac{365}{11}$

36) d)

$-\frac{32}{11}z^3+5z^2+12z-\frac{365}{11}=0$

$32z^3-55z^2-132z+365=0$

Since we know $p\alpha^3+5\alpha^2+12\alpha+q=0$ has solution $\alpha=2-\mathrm{i}$ for these values of p and q and that the coefficients are real two of the roots are $2-\mathrm{i}$ and $2+\mathrm{i}$

$\big(z-(2-\mathrm{i})\big)\big(z-(2+\mathrm{i})\big)=z^2-4z+5$

$32z^3-55z^2-132z+365=(z^2-4z+5)(32z+73)$

So the solutions are $z=2-\mathrm{i}$, $z=2+\mathrm{i}$ and $z=-\frac{73}{32}$

37) a)

$z^2 - 3z + 9 = 0$ has roots $z = \frac{3}{2} \pm \frac{3\sqrt{3}}{2}\mathrm{i}$

$r = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{3\sqrt{3}}{2}\right)^2} = 3$

$\theta = \tan^{-1}\left(\frac{\frac{3\sqrt{3}}{2}}{\frac{3}{2}}\right) = \frac{\pi}{3}$

So in modulus-argument form the roots are:

$3\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$ and $3\left(\cos\frac{\pi}{3} - \mathrm{i}\sin\frac{\pi}{3}\right)$

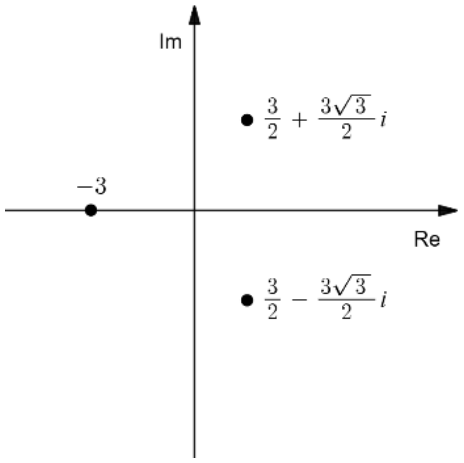
37) b)

$z^3 + k = (z^2 - 3z + 9)(z + 3)$

So $k = 27$

The final root is $z = -3$

37) c)



38) a)

By inspection,

$1 + k - k - 1 = 0$

$\therefore z = 1$ is a root of the equation.

38) b)

	z^2	$(1+k)z$	$+1$
z	z^3	$(1+k)z^2$	$+z$
-1	$-z^2$	$-(1+k)z$	-1

$\therefore z^3 + kz^2 - kz - 1 = (z-1)(z^2 + (1+k)z + 1)$

For $z^2 + (1+k)z + 1$ to have complex roots $b^2 - 4ac < 0$

$(1+k)^2 - 4 \times 1 \times 1 < 0$

$k^2 + 2k - 3 < 0$

$(k+3)(k-1) < 0$

$\therefore -3 < k < 1$

39) a)

Two of the roots are $5 + 2\text{i}$ and $5 - 2\text{i}$

$\big(z - (5+2\text{i})\big)\big(z - (5-2\text{i})\big) = z^2 - 10z + 29$

$\therefore z^2 - 10z + 29$ is a factor of $\text{f}(x)$

39) b)

	z^4	$+0z^3$	$+0z^2$	$+0z$	-16
z^2	z^6	$+0z^5$	$+0z^4$	$+0z^3$	$-16z^2$
$-10z$	$-10z^5$	$-0z^4$	$-0z^3$	$-0z^2$	$+160z$
$+29$	$+29z^4$	$+0z^3$	$+0z^2$	$+0z$	-464

So $z^4 - 16$ is also a factor

By inspection:

$z^4 - 16 = 0$ has solutions 2, -2, 2i and -2i

So all the solutions are $5 + 2\text{i}$, $5 - 2\text{i}$, 2, -2, 2i and -2i

40)

Two of the solutions are $4-7\mathrm{i}$ and $4+7\mathrm{i}$

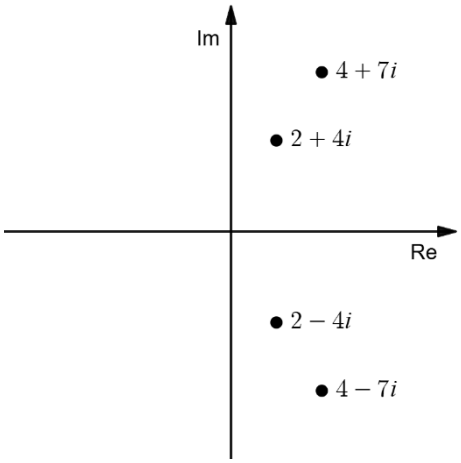
$\big(z-(4-7\mathrm{i})\big)\big(z-(4+7\mathrm{i})\big)=z^2-8z+65$

	z^2	$-4z$	$+20$
z^2	z^4	$-4z^3$	$20z^2$
$-8z$	$-8z^3$	$+32z^2$	$-160z$
$+65$	$+65z^2$	$-260z$	$+1300$

And

$z^2-4z+20=0$ has solutions $z=2-4\mathrm{i}$ and $z=2+4\mathrm{i}$

$\therefore$ the roots of $z^4-12z^3+117z^2-420z+1300=0$ are $4-7\mathrm{i},\ 4+7\mathrm{i}, 2-4\mathrm{i}$ and $2+4\mathrm{i}$



Trapezium area $=\frac{1}{2}(8+14)\times 2=22$ units^2

41)

One root must be real.

Let $\mathrm{f}(z)=z^3-12z^2+47z-54$

$\mathrm{f}(1)=1-12+47-54\neq 0$

$\mathrm{f}(2)=8-48+94-54=0\ \therefore (z-2)$ is a factor

$(z-2)(z^2-10z+27)=0$

So the roots are $z=2$, $z=5+\sqrt{2}\mathrm{i}$ and $z=5-\sqrt{2}\mathrm{i}$

Area is $\frac{1}{2}\times 3\times 2\sqrt{2}=3\sqrt{2}$ units^2.

42) a)

Area is 64 units^2

42) b)

Two of the roots are $3 + 4\text{i}$ and $3 - 4\text{i}$

The other roots to form the square are $-5 + 4\text{i}$ and $-5 - 4\text{i}$

So the polynomial is:

$\big(z - (3 + 4\text{i})\big)\big(z - (3 - 4\text{i})\big)\big(z - (-5 + 4\text{i})\big)\big(z - (-5 - 4\text{i})\big)$

$= (z^2 - 6z + 25)(z^2 + 10z + 41)$

$= z^4 + 10z^3 + 41z^2 - 6z^3 - 60z^2 - 246z + 25z^2 + 250z + 1025$

$= z^4 + 4z^3 + 6z^2 + 4z + 1025$

So $a = 4$, $b = 6$, $c = 4$ and $d = 1025$

43)

Let $\text{f}(z) = z^3 - 7z^2 + az + b$

Let the complex roots be $p + q\text{i}$ and $p - q\text{i}$

$\text{f}(z) = (z - 3)\big(z - (p + q\text{i})\big)\big(z - (p - q\text{i})\big)$

$= (z - 3)(z^2 - 2pz + p^2 + q^2)$

$= z^3 - 2pz^2 + (p^2 + q^2)z - 3z^2 + 6pz - 3(p^2 + q^2)$

$= z^3 - (2p + 3)z^2 + (p^2 + 6p + q^2)z - 3(p^2 + q^2)$

So $2p + 3 = 7 \therefore p = 2$

Area is $56 = \frac{1}{2} \times (2 - 3)(2q)$ which gives $q = -56$

$p = 2$ and $q = -56$

$\therefore z^3 - (4 + 3)z^2 + (4 + 12 + 3136)z - 3(4 + 3136)$

$= z^3 - 7z^2 + 3152z - 9420$

$\therefore a = 3152$ and $b = -9420$

44)

$\big(z - (a + b\text{i})\big)\big(z - (a - b\text{i})\big)(z - k) = (z^2 - 2az + a^2 + b^2)(z - k)$

$= z^3 - 2az^2 + (a^2 + b^2)z - kz^2 + 2kaz - k(a^2 + b^2)$

$= z^3 - (2a + k)z^2 + (a^2 + 2ka + b^2)z - k(a^2 + b^2)$

$\text{f}(z) = z^3 - (2a + k)z^2 + (a^2 + 2ka + b^2)z - k(a^2 + b^2)$

45)

One of the roots must be real.

Let $f(x) = z^3 - 23z^2 + 299z - 2197$

By testing values of x which are factors of -2197:

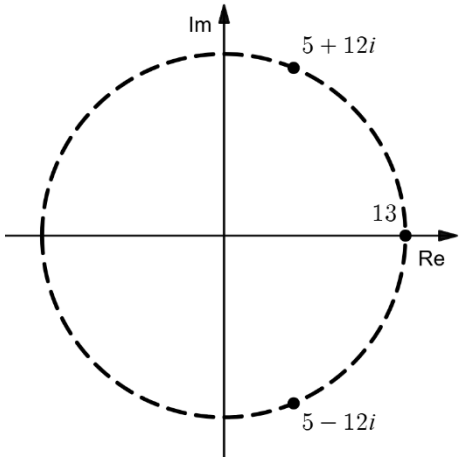
$f(13) = z^3 - 23z^2 + 299z - 2197 = 2197 - 3887 + 3887 - 2197 = 0$

$\therefore (z - 13)$ is a factor of $z^3 - 23z^2 + 299z - 2197$

	z^2	$-10z$	$+169$
z	z^3	$-10z^2$	$+169z$
-13	$-13z^2$	$+130z$	-2197

and

$z^2 - 10z + 169 = 0$ has solutions $z = 5 + 12\text{i}$ and $z = 5 - 12\text{i}$



Distance to each point from the origin:

$\sqrt{5^2 + 12^2} = 13$, $\sqrt{5^2 + (-12)^2} = 13$ and 13 so they all lie on a circle of radius 13 centred at the origin. So that $|z| = 13$

46)

Complex roots have a distance of 6 between them so they are of the form

$p + 3\text{i}$ and $p - 3\text{i}$

$5 = \sqrt{p^2 + 3^2} \Rightarrow p = 4$

Area is $42 = \frac{1}{2} \times 6 \times h$ so $h = 14$

So possible roots are $4 + 3\text{i}$, $4 - 3\text{i}$ and 18 or $4 + 3\text{i}$, $4 - 3\text{i}$ and -10

So the equations are:

$\big(z - (4 + 3\text{i})\big)\big(z - (4 - 3\text{i})\big)(z - 18) = (z^2 - 8z + 25)(z - 18)$

$= z^3 - 26z^2 + 169z - 450$

$\big(z - (4 + 3\text{i})\big)\big(z - (4 - 3\text{i})\big)(z + 10) = (z^2 - 8z + 25)(z + 10)$

$= z^3 + 2z^2 - 55z + 250$

So the equations are $\text{f}(z) = z^3 - 26z^2 + 169z - 450$ or $\text{f}(z) = z^3 + 2z^2 - 55z + 250$

47)

Since $x(x^3 - x^2 + 8x + 10) = 0$

Let $\text{f}(x) = x^3 - x^2 + 8x + 10$

$\text{f}(-1) = -1 - 1 - 8 + 10 = 0$

So $x = -1$ is a factor

So $x(x + 1)(x^2 - 2x + 10) = 0$ has solutions 0, -1, $1 + 3\text{i}$ and $1 - 3\text{i}$

$\therefore a = -1$

So when $a = -1$, -2i is a root of

$z^4 - 3z^2 + 14z - 28 = 0$

Since $(z - 2\text{i})(z + 2\text{i}) = z^2 + 4$

$z^4 - 3z^2 - 28 = (z^2 + 4)(z^2 - 7)$

$\therefore$ the values of z are $z = 2\text{i}, z = -2\text{i}, z = \sqrt{7}$ and $z = -\sqrt{7}$

48) a)

$a + 3 - 3\text{i}$

48) b)

$(z-b)(z-a)\big(z-(a+3+3\mathrm{i})\big)\big(z-(a+3-3\mathrm{i})\big)$

$=(z-b)(z-a)(z^2-(2a+6)z+a^2+6a+18)$

$=(z^2-(a+b)z+ab)(z^2-(2a+6)z+a^2+6a+18)$

Compare cubic coefficients:

$-(a+b)-(2a+6)=-8 \Rightarrow b+3a=2 \Rightarrow b=2-3a$

Compare the quadratic coefficients:

$(a+b)(2a+6)+ab+(a^2+6a+18)=6$

Substitute in $b=2-3a$ and solve:

$(a+2-3a)(2a+6)+a(2-3a)+(a^2+6a+18)=6$

$(2-2a)(2a+6)+2a-3a^2+a^2+6a+18=6$

$-4a^2-12a+4a+12+2a-3a^2+a^2+6a+18=6$

$-6a^2+24=0$

$a^2=4$

$\therefore a=2$ or $a=-2$ But since $a>0$

$a=2$

$a=2$ gives $b=-4$

$\therefore a=2$ and $b=-4$

48) c)

$\therefore$ the solutions are $z=2, z=-4, z=5-3\mathrm{i}$ and $z=5+3\mathrm{i}$

49) a)

Two of the solutions are $5+\mathrm{i}$ and $5-\mathrm{i}$

$\big(z-(5+\mathrm{i})\big)\big(z-(5-\mathrm{i})\big)=z^2-10z+26$

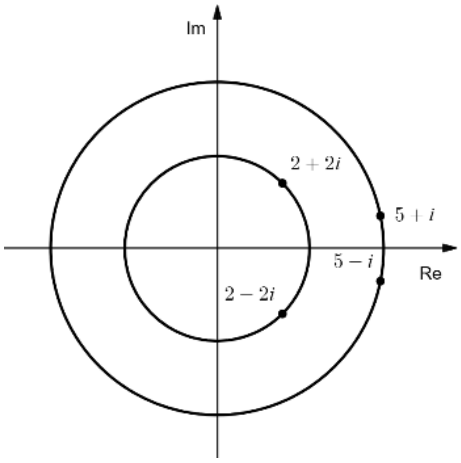
	z^2	$-4z$	$+8$
z^2	z^4	$-4z^3$	$+8z^2$
$-10z$	$-10z^3$	$+40z^2$	$-80z$
$+26$	$+26z^2$	$-104z$	$+208$

And

$z^2-4z+8=0$ has solutions $z=2-2\mathrm{i}$ and $z=2+2\mathrm{i}$

$\therefore$ the roots of $z^4-14z^3+74z^2-184z+208=0$ are $5+\mathrm{i},\ 5-\mathrm{i}, 2-2\mathrm{i}$ and $2+2\mathrm{i}$

49) b)



Radius of larger circle: $\sqrt{5^2+1^2}=\sqrt{26}$
Radius of smaller circle: $\sqrt{2^2+2^2}=\sqrt{8}$
Area is $\pi \times 26-\pi \times 8=18\pi$ units^2

50) a)

$z^2-1=0$ has roots $z=1$ and $z=-1$
$1-1=0$

50) b)

$z^3-1=0$
$(z-1)(z^2+z+1)=0$
has roots $z=1$, $z=-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$ and $z=-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$
$1-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}=0$

50) c)

$z^4-1=0$
$(z^2-1)(z^2+1)=0$
has roots $z=1$, $z=-1$, $z=\mathrm{i}$ and $z=-\mathrm{i}$
$1-1+\mathrm{i}-\mathrm{i}=0$

51) a)

Since $z = p + q\text{i}$ and $z = p - q\text{i}$ is a root:

$pz^3 + qz^2 + qz + p = (z - k)(p + q\text{i})(p - q\text{i})$

$= (z^2 - 2pz + p^2 + q^2)(z - k)$

$= z^3 - 2pz^2 + (p^2 + q^2)z - kz^2 + 2kpz - k(p^2 + q^2)$

$= z^3 - (2p + k)z^2 + (p^2 + 2kp + q^2)z - k(p^2 + q^2)$

Since the linear term is equal to the quadratic term:

$p^2 + 2kp + q^2 = -(2p + k)$

$p^2 + q^2 + 2p = -2kp - k$

$p^2 + q^2 + 2p = -k(2p + 1)$

So $k = -\frac{p^2+q^2+2p}{2p+1}$ as req.

51) b)

$q = 0$ so

$k = -\frac{p^2+2p}{2p+1} = -\frac{p(p+2)}{2p+1}$

52)

Assume that it has two complex roots, $p + q\text{i}$ and $p - q\text{i}$

So that

$\big(z - (p + q\text{i})\big)\big(z - (p - q\text{i})\big) = z^2 - 2pz + p^2 + q^2$

Assume a real root exists, k so that

$\text{f}(z) = \big(z - (p + q\text{i})\big)\big(z - (p - q\text{i})\big)(z - k)$

$= (z^2 - 2pz + p^2 + q^2)(z - k)$

$= z^3 - 2pz^2 + (p^2 + q^2)z - kz^2 + 2pkz - k(p^2 + q^2)$

$= z^3 - (2p + k)z^2 + (p^2 + 2pk + q^2)z - k(p^2 + q^2)$

Since

$p^2 + 2pk + q^2 = 0$

For p to be real then the discriminant of $p^2 + 2pk + q^2$ must be ≥ 0

$\therefore (2k)^2 - 4q^2 \geq 0$

$4k^2 - 4q^2 \geq 0$

$k^2 \geq q^2$

53)

Assume that it has two complex roots, $p+q\mathrm{i}$ and $p-q\mathrm{i}$

So that

$\big(z-(p+q\mathrm{i})\big)\big(z-(p-q\mathrm{i})\big)=z^2-2pz+p^2+q^2$

Assume a two real roots exist, m and n so that

$\mathrm{f}(z)=\big(z-(p+q\mathrm{i})\big)\big(z-(p-q\mathrm{i})\big)(z-m)(z-n)$

$=(z^2-2pz+p^2+q^2)(z^2-(m+n)z+mn)$

$=z^4-(m+n)z^3+mnz^2-2pz^3+2p(m+n)z^2-2pmnz+(p^2+q^2)z^2-$

$(p^2+q^2)(m+n)z+mn(p^2+q^2)$

$=z^4-(m+n+2p)z^3+(2p(m+n)+mn+(p^2+q^2))z^2-\big(2pmn+$

$(p^2+q^2)(m+n)\big)z+mn(p^2+q^2)$

Since

$2pmn+(p^2+q^2)(m+n)=0$

$(m+n)p^2+2mnp+(m+n)q^2=0$

For p to be real then the discriminant of $(m+n)p^2+2mnp+(m+n)q^2$

must be ≥ 0

$\therefore (2mn)^2-4(m+n)(m+n)q^2\geq 0$

$4(mn)^2\geq 4(m+n)^2q^2$

$(mn)^2\geq (m+n)^2q^2$

$\frac{(mn)^2}{(m+n)^2}\geq q^2$

Since m, n and q are real and non-zero then

$\frac{mn}{m+n}\geq q$ as req.